



Explore cloud deployment models

Welcome to this intro to cloud deployment models!

We're going to explore how to advise any organization in choosing the right environment for their unique business needs.

There are three primary models: public clouds, private clouds, and hybrid clouds.

As a cloud data analytics professional, it will be important to understand how each works.

Then, you can help your organization select a model that's flexible, adaptable, and helps users quickly respond and adjust to changing conditions.

First up, a public cloud is a cloud model that delivers computing, storage, and network resources through the internet.

In this model, these resources are shared among multiple users and organizations, granting them on-demand access and utilization.

Public cloud services are overseen and maintained by third-party cloud service providers, who not only manage the infrastructure but also operate their own data centers.

Next, a private cloud is a cloud model that dedicates all cloud resources to a single user or organization and is created, managed, and owned within on-premises data centers.

Finally, hybrid clouds are a combination of the public and private models.

They enable organizations to enjoy both cloud services and the control features of on-premises cloud models.

Think of these three cloud models as different ways to heat a building.

Public clouds are like an electricity company that delivers the power you use to generate heat.

You can choose to turn it up when you're chilly or turn it off when it's warm outside.

You only pay for the power used.

And, as a customer, you don't need to worry about the maintenance of the power lines and generators.

Private clouds are like having your own solar panels to generate power and heat.

You need to buy the panels, have a place to install them, and you're responsible for their proper care and maintenance.

And hybrid clouds are like using the services of an electricity company, but also owning and using solar panels.

This gives you more options over how heat is delivered when the temperatures drop.

You can choose when the power company is right to use and when solar panels are the better option.

Ok, now let's discuss the **advantages and disadvantages** of each model.

With **public clouds**, you **pick and choose the resources** you need, and **pay only for what you use**.

Public clouds can **easily scale up or down** based **on demand**.

And there are **no maintenance** worries because the cloud service provider handles all of that for you.

Another key advantage is **reliability**.

Public clouds have **vast networks of servers** and can **quickly redirect resources in an emergency**.

Speed and ease of deployment are also benefits of a public cloud model.

Adoption occurs faster and more simply because the cloud infrastructure is already in place.

Lastly, public clouds **offer new services and frequent updates** that enable users to benefit from the latest innovations like artificial intelligence and machine learning, or **AI and ML**.

Now, **private clouds** come with **higher maintenance and management costs**, but offer a few critical advantages.

The first is that, as the name suggests, they **offer private and secure networks** if **protected properly**.

Without proper security measures put in place, it can be **vulnerable to hackers**.

Second, **private clouds help** with any required **regulations and compliance** because **you control where your data is stored and where computing takes place**.

Private clouds also **provide consistent performance** because **hardware isn't shared among other organizations**.

Going hybrid can be a bit **tricky**, as blending **public and private models adds complexity**.

But there are some key benefits.

A hybrid cloud model allows you to add a public cloud provider to your existing on-premises infrastructure, which increases cloud computing power without adding data center expenses.

A hybrid cloud model also gives you access to the latest innovations, like AI and ML, which can really be business game changers.

Because you choose where your applications sit and where computing happens, there are key security and compliance advantages.

Likewise, hybrid computing occurs closer to the actual users, so it enables faster performance.

There's also greater flexibility because you can choose whichever cloud environment is best for the specific task at hand.

Whether your organization is best suited to a public, private, or hybrid cloud model, your guidance will help choose a great option.

All three offer some really cool ways to advance any organization's computing power!